

Module 4 – DBMS & SQL Lab Answers

Create Database:

```
CREATE DATABASE school_db;
```

Create Students Table:

```
CREATE TABLE students(student_id INT PRIMARY KEY, student_name VARCHAR(50), age INT, class VARCHAR(20), address VARCHAR(100));
```

Insert Data:

```
INSERT INTO students VALUES (1,'A',12,'7A','City');
```

```
SELECT * FROM students;
```

Select Specific Columns:

```
SELECT student_name, age FROM students;
```

Condition Query:

```
SELECT * FROM students WHERE age > 10;
```

Create Teachers Table:

```
CREATE TABLE teachers(teacher_id INT PRIMARY KEY, teacher_name VARCHAR(50) NOT NULL, subject VARCHAR(50) NOT NULL, email VARCHAR(100) UNIQUE);
```

Foreign Key:

```
ALTER TABLE students ADD teacher_id INT;
```

```
ALTER TABLE students ADD FOREIGN KEY (teacher_id) REFERENCES teachers(teacher_id);
```

Courses Table:

```
CREATE TABLE courses(course_id INT PRIMARY KEY, course_name VARCHAR(50), course_credits INT);
```

ALTER Table:

```
ALTER TABLE courses ADD course_duration INT;
```

```
ALTER TABLE courses DROP COLUMN course_credits;
```

DROP Table:

```
DROP TABLE teachers;
```

DML:

```
INSERT INTO courses VALUES (1,'Java',3);
```

```
UPDATE courses SET course_duration=6 WHERE course_id=1;
```

```
DELETE FROM courses WHERE course_id=1;
```

DQL:

```
SELECT * FROM courses;
```

```
SELECT * FROM courses ORDER BY course_duration DESC;
```

```
SELECT * FROM courses LIMIT 2;
```

DCL:

```
GRANT SELECT ON courses TO user1;
```

```
REVOKE INSERT ON courses FROM user1;
```

TCL:

```
COMMIT;
```

```
ROLLBACK;
```

JOINS:

```
SELECT * FROM employees INNER JOIN departments ON  
employees.dept_id=departments.dept_id;
```

GROUP BY:

```
SELECT dept_id, COUNT(*) FROM employees GROUP BY dept_id;
```

Stored Procedure:

```
CREATE PROCEDURE get_emp() BEGIN SELECT * FROM employees; END;
```

View:

```
CREATE VIEW emp_view AS SELECT * FROM employees;
```

Trigger:

```
CREATE TRIGGER trg AFTER INSERT ON employees FOR EACH ROW INSERT INTO log  
VALUES ('Inserted');
```

PLSQL:

```
BEGIN DBMS_OUTPUT.PUT_LINE('Hello'); END;
```

Cursor:

```
DECLARE CURSOR c IS SELECT * FROM employees;
```

SAVEPOINT:

```
SAVEPOINT sp1;
```